Practice Problems on Fatou's Lemma

Problem 1: Direct computation

Let $f_n(x)=x^n$ on $[0,1]$. Use Fatou's lemma to compare $\int \liminf f_n$ and $\liminf \int f_n$. Compute both sides explicitly.

Problem 2: A sequence converging to infinity

On $(0,1]$, define $f_n(x)=n$ for $0<x<1/n$ and 0 otherwise. Compute $\liminf f_n(x)$ and verify Fatou's lemma.

Problem 3: Moving intervals

Let $f_n(x)=1_{[n,n+1]}(x)$ on $\mathbb{R}$. Compute $\int \liminf f_n$ and $\liminf \int f_n$. Determine whether equality holds.

Problem 4: Showing a limit is integrable

Suppose $f_n\geq 0$, $\int_0^1 f_n \leq 5$ for every n, and $f_n\to f$ a.e. Use Fatou's lemma to prove $\int_0^1 f \leq 5$.

Problem 5: Expectation version

Let $X_n\geq 0$ be random variables with $X_n\to X$ almost surely and $E[X_n]\leq 10$. Use Fatou's lemma to show $E[X]\leq 10$.

Problem 6: Strict inequality

Using the sequence from Problem 2, show that $\int \liminf f_n < \liminf \int f_n$.

Problem 7: Recovering MCT

Suppose $0\leq f_1\leq f_2\leq \dots$ and $f_n\uparrow f$. Use Fatou's lemma and monotonicity of the integral to prove $\int f = \lim \int f_n$.

Problem 8: Bounded second moments

Let $X_n\to X$ almost surely and $E[X_n^2]\leq 4$ for all n. Use Fatou's lemma to prove $E[X^2]\leq 4$.

Problem 9: An L1 application

Let $f_n\geq 0$ satisfy $\int_0^\infty f_n \leq M$ for all n and $f_n\to f$ a.e. Prove that f belongs to $L1([0,\infty))$.

Problem 10: Standard proof of Fatou's lemma

Let $g=\liminf f_n$ and define $g_k=\inf_{n\geq k} f_n$. Show (i) g_k is increasing; (ii) $g_k\uparrow g$; (iii) $\int g_k \leq \inf_{n\geq k}\int f_n$; then complete the proof of Fatou's lemma.

Challenge Problem

Construct a sequence $f_n \geq 0$ such that $\int \liminf f_n = 0$ but $\liminf \int f_n = 1$.